

32
50

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DSA 0304

1. B) Recursive Production Rules allow hierarchical sentence Generation
2. A) subject - verb agreement
3. X) Finite State Automata
4. X) Morphological Analyzer
5. B) left recursion causing infinite expansion.
6. B) uses predictor, scanner, and completer operations to process partial input efficiently
7. X) Regular expression matching
8. A) choose the parse tree with the maximum rule probability
9. C) Finite state transducer + forward scanner
10. A) CFG + Feature structure
11. X) Remove all PP values $NP \leftarrow NP.PP$
12. X) the list of ~~best~~ authorized employees are updated
13. C) the verb relates its subcategorical frame.
14. B) left recursion ~~is~~ causing infinite recursion
15. D) dynamic chart construction combining predictor, scanner and completer.

16. ☒ A) Number feature conflict between subject and verb
17. ☒ A) Tree1 because it has high probability.
18. ☒ A) Highly recursive grammar with unrestricted predictions
19. ☒ D) Early parsing handles ambiguity efficiently, while PCFG
20. ☒ C) CFG parsing $\rightarrow$ Subcategorization checking $\rightarrow$ Feature
21. ☒ C) Introduce a compound noun production $N \rightarrow$ 'living room' N
22. ☒ A) main subject noun phrase
23. ☒ D) Verb subcategorization frame
24. ☒ A) Shift reduce conflict
25. ☒ C) completer
26. ☒ B) maximum likelihood estimation
27. ☒ A) Tree A because it has the highest probability
28. ☒ A) Context-free Grammar only
29. ☒ B) Recursive top-down parser
30. ☒ A) Maximum frequency matching
31. ☒ A) $S \rightarrow NP VP \rightarrow Det N VP \rightarrow$ the robot $VP \rightarrow V NP \rightarrow$ delivers $Det N \rightarrow$ the package
32. ☒ C) The engineers developed software
33. ☒ C) She gives the customer a refund
34. ☒ D) Expand s into NP VP
35. ☒ C) completer
36. ☒ B) Unification fails because feature value conflict
37. ☒ A) The phrase will be the highest overall probability
38. ☒ B) Recursive decent parsing

40. ☒ A) False A because it satisfies the constraints and has the highest probability
41. ☒ B) Reject the False because grammatical agreement constraints are violated despite syntactic correctness
42. ☒ C) S-NPVP
43. ☒ D) VP
44. ☒ A) Predictor
45. ☒ A) sentence accepted because PERSON matches
46. ☒ A) The maximum number of production rules
47. ☒ A) CFG suggests recursive noun phrase generation and
 $NP \rightarrow Adj. NP$
48. ☒ C) Structural (syntactic) ambiguity
49. ☒ X) recursive grammar expansion
50. ☒ A) The verb "gave" requires additional complement
51. ☒ C) V NP PP